ITEM	QTY	CATALOG	MANUFACTURER	DESCRIPTION	INSTRUCTIONS
1	1	ACE - PROTECTION SHIELD	ACE MANUFACTURING	CUSTOM PANEL PROTECTION SHIELD, SEE DRAWINGS FOR DIMENSIONS AND DETAILS	SEE NOTE 2
2	1	RTD1-B01L06-01	AUTOMATION DIRECT	TEMPERATURE SENSOR, PT100 RTD, BOLT-ON RING, BRASS SHEATH, #6-#10 (4MM-5MM) SCREW OR BOLT SIZE PROCESS CONNECTION, 6FT 24 AWG WIRE LEADS	
3	4	KTK-R-15A	BUSSMANN	FUSE KTK-R BRANCH CIRCUIT 15 AMP	
4	2	KTK-R-20A	BUSSMANN	FUSE KTK-R BRANCH CIRCUIT 20 AMP	
5	2	MD3C2-MTRJSC-3	CIRCAMAX	MULTIMODE (OM3) 2mm DUPLEX MTRJ-SC 50/125 FIBER PATCH CABLE, 3 FEET	
6	1	MD3C2-SCSC-3	CIRCAMAX	MULTIMODE (OM3) 2mm DUPLEX SC-SC 50/125 FIBER PATCH CABLE, 3 FEET	
7	1	SNAP-12SC-MM OM3	DINSPACE	SNAP COMPACT FIBER OPTIC PATCH PANEL W/6 EA MULTI-MODE OM3 SC/SC DUPLEX ADAPTERS	
8	2	S501-10-R	EATON	FUSE CERAMIC S501 5x20mm 10 AMP	
9	8	S501-2-R	EATON	FUSE CERAMIC S501 5x20mm 2 AMP	
10	2	S501-5-R	EATON	FUSE CERAMIC S501 5x20mm 5 AMP	
11	4	S501-6.3-R	EATON	FUSE CERAMIC S501 5x20mm 6.3 AMP	
12	1	*DISTRIBUTED CHARM ASSEMBLY	EMERSON	DISTRIBUTED CHARM ASSEMBLIES	SEE NOTE 1
	1	SE4701T02	EMERSON	12 CHARM I/O JUNCTION BOX WITH 1/2 INCH - 14 NPT THREADS AND REDUNDANT CHARM I/O BLOCK ASSEMBLY	
	0	SE4301T02	EMERSON	DISCRETE INPUT CHARM; 24 VDC DRY CONTACT; LOW-SIDE SENSE	
	3	SE4302T02	EMERSON	DISCRETE OUTPUT CHARM; 24 VDC; ISOLATED	
	6	SE4303T01	EMERSON	ANALOG INPUT CHARM; 4-20 MA; HART	
	3	SE4304T01	EMERSON	ANALOG OUTPUT CHARM; 4-20 MA; HART	
	0	SE4302T05	EMERSON	24 VDC POWER CHARM	
	0	SE6103-SPEC	EMERSON	PROTECTION COVER FOR CHARMS (SINGLE)	
	3	SE4502	EMERSON	FUSED-INJECTED-POWER CHARM TERMINAL BLOCK	
13	3	--CHARM BASEPLATES--	EMERSON	CHARM BASEPLATE ASSEMBLIES; SEE FOLLOWING LINES FOR DETAILS	
	2	SE4601T07	EMERSON	CHARMS BASEPLATE ASSEMBLY, INCLUDES CHARMS BASEPLATE, CHARM ADDRESS TERMINAL BLOCK WITH SCREW TERMINALS,12 STANDARD CHARM TERMINAL BLOCKS AND A BASEPLATE IDENTIFIER	
	1	SE4601T08	EMERSON	CHARMS BASEPLATE ASSEMBLY WITH FUSED-INJECTED FIELD POWER, INCLUDES CHARMS BASEPLATE, CHARM ADDRESS TERMINAL BLOCK WITH SCREW TERMINALS, 12 FUSED-INJECTED-POWER CHARM TERMINAL BLOCKS AND A BASEPLATE IDENTIFIER	
14	36	--CHARM MODULES--	EMERSON	CHARM MODULES; SEE FOLLOWING LINES FOR DETAILS	
	11	SE4301T02	EMERSON	DISCRETE INPUT CHARM; 24 VDC DRY CONTACT; LOW-SIDE SENSE	
	9	SE4302T02	EMERSON	DISCRETE OUTPUT CHARM; 24 VDC; ISOLATED	
	6	SE4303T01	EMERSON	ANALOG INPUT CHARM; 4-20 MA; HART	
	3	SE4304T01	EMERSON	ANALOG OUTPUT CHARM; 4-20 MA; HART	
	1	SE4303T03	EMERSON	RTD INPUT CHARM	
	6	SE6103-SPEC	EMERSON	PROTECTION COVER FOR CHARMS (SINGLE)	
15	1	SE4603T05	EMERSON	CHARM BASEPLATE EXTENDER WITH CABLE CONNECTORS TOP (MALE) (INCL IDENTIFIER & SEPARATION WALL)	
16	1	SE4603T06	EMERSON	CHARM BASEPLATE EXTENDER WITH CABLE CONNECTORS BOTTOM (FEMALE) (INCL IDENTIFIER & SEPARATION WALL)	
17	1	SE4604T04	EMERSON	CHARM BASEPLATE TERMINATOR, BOTTOM (FEMALE) (INCL IDENTIFIER & SEPARATION WALL)	
18	1	SE4605T03	EMERSON	CHARMS BASEPLATE CABLES, INCLUDES TWO 1.1M CABLES	
19	1	SE4606T02	EMERSON	CHANNEL IDENTIFIER LABELS FOR CHARM BASEPLATES (PKG OF 8)	
20	1	SE4613	EMERSON	CHARMS ADDRESS PLUGS 1 THROUGH 4	
21	1	SE6505T02	EMERSON	REDUNDANT CIOC2 WITH FIBER ETHERNET, INCLUDES REDUNDANT PAIR OF CHARM I/O CARD 2, CHARM I/O CARRIER	
22	1	SE6701T02	EMERSON	REDUNDANT CHARM I/O GATEWAY ASSEMBLY FOR RING TOPOLOGY. INCLUDES GATEWAY CARRIER, 2 SCANNER MODULES, 2, 1-PORT ETHERNET IOPS	
23	4	VE4054DRS-SPEC	EMERSON	DELTAV DIN-RAIL STOP (QTY 1)	
24	2	VE5126	EMERSON	BULK POWER SUPPLY 100-240VAC TO 24VDC, 20A	
25	1	VE5135	EMERSON	POWER SUPPLY REDUNDANCY MODULE 12-28V, 40A	
26	2	2907918	PHOENIX CONTACT	PLT-SEC-T3-120-FM-UT TYPE 3 SURGE PROTECTION DEVICE C/W PLT-SEC-T3-120-P-UT/PT PLUG (2907922) & PLT-SEC-T3-BE-FM-UT BASE (2907924)	
27	1	106FX2-SC	RED LION	N-TRON 106FX UNMANAGED INDUSTRIAL ETHERNET SWITCH, 4 10/100BASE TX RJ-45 PORTS, 2 100BASEFX MM PORT WITH SC CONNECTORS. CUL CLASS 1 DIV 2	
28	1	DPE4004S3	REDAPT	BREATHER/DRAIN M20 GF NYLON TYPE 4X C1Z1 - NITRIL O-RING	
29	1	2513.000	RITTAL	PRINT POCKET A3 11"(286mm)H X 17"(438mmW) X 1.8"(45mm)D PLASTIC LANDSCAPE	
30	1	8017.601	RITTAL	WM483612N4 TYPE 4X WALLMOUNT CABINET, 1200mmH X 900mmW X 300mmD, 304 SS C/W ZINC MTG PLATE, LHS HINGED DOOR AND 3 POINT LATCHING L-HANDLE	
31	1	8018.791	RITTAL	WALL MOUNTING BRACKETS STAINLESS STEEL 316L WMSMB FOR USE WITH KL, EB/JB, AND AE/WM ENCLOSURES	
32	1	70-750DL11-1	SCHNEIDER	RELAY SOCKETS & HARDWARE DINPM OCTAL SOCKET 11-PIN, LOGIC STYLE	
	1	750XCXH-24D	SCHNEIDER	750H HERMETIC OCTAL RELAY, 11 PIN/3PDT, 12AMP RATING	
	1	16-1351-SPEC	SCHNEIDER	HOLD DOWN CLIP (QTY 1)	
33	1	70-750DL11-1	SCHNEIDER	RELAY SOCKETS & HARDWARE DINPM OCTAL SOCKET 11-PIN, LOGIC STYLE	
	1	750XCXH-120A	SCHNEIDER	POWER RELAY, SE RELAYS, 12A, 3PDT, 120 VAC, HERMETICALLY SEALED CONTACTS, HAZARDOUS LOCATION SERIES	
	1	16-1351-SPEC	SCHNEIDER	HOLD DOWN CLIP (QTY 1)	
34	1	-- CGB ASSEMBLY --	SPARTAN CONTROLS	CHASSIS GROUND BAR ASSEMBLY - SEE FOLLOWING LINES FOR DETAILS	
	1	0003111	ALIX MACHINING	1/4" BARE COPPER BUSS BAR; 6"L X 1 1/2"W	
	1	TA-2/0I	ILSCO	GROUND LUG, 2/0 AWG - #6 AWG (70 mm² - 16 mm²) (GROUND LUG MEDIUM)	

ITEM	QTY	CATALOG	MANUFACTURER	DESCRIPTION	INSTRUCTIONS
35	1	-- DCGB ASSEMBLY --	SPARTAN CONTROLS	DC GROUND BAR ASSEMBLY - SEE FOLLOWING LINES FOR DETAILS	
	1	0003111	ALIX MACHINING	1/4" BARE COPPER BUSS BAR; 6"L X 1 1/2"W	
	2	ISO1-14	ERIFLEX	1" GLASS FIBRE REINFORCED POLYESTER STEEL INSULATOR	
	1	TA-2/0I	ILSCO	GROUND LUG, 2/0 AWG - #6 AWG (70 mm² - 16 mm²) (GROUND LUG MEDIUM)	
36	1	--GROUND BAR (12")--	SPARTAN CONTROLS	GROUND BAR ASSEMBLY, ALUMINUM, 38 USEABLE POINTS	
	1	0000075	ILSCO	N8020 ALUMINUM BUS BAR (12" LENGTH)	
37	1	0000094B	SPARTAN CONTROLS	LOGO SPARTAN METAL (COLORED)	
38	16	1011300001	WEIDMULLER	FUSED TERMINAL BLOCK WSI 6/LD 10-36 VAC/DC 0.5MA LED LOW CURRENT	
39	30	1020100000	WEIDMULLER	TERMINAL BLOCK WDU 4 FEED-THROUGH	
40	3	1020200000	WEIDMULLER	TERMINAL BLOCK WDU 6 FEED-THROUGH	
41	10	1020300000	WEIDMULLER	TERMINAL BLOCK WDU 10 FEED-THROUGH	
42	10	1023300000	WEIDMULLER	TERMINAL BLOCK WDK 2.5 DOUBLE-TIER, FEED-THROUGH, SCREW CONNECTION W/ 1 DIODE A.2	
43	23	1050000000	WEIDMULLER	END PLATE WAP 2.5-10/DARK BEIGE	
44	34	1061200000	WEIDMULLER	END BRACKET WEW 35/2	
45	6	1966060000	WEIDMULLER	FUSED TERMINAL BLOCK WSI 25/1 10X38/LED	
46	1	7914760001	WEIDMULLER	UTILITY FUSE BOX PLASTIC DIN RAIL MOUNT	SEE NOTE 3

SPARE FUSES

QTY	CATALOG	MANUFACTURER	DESCRIPTION	INSTRUCTIONS
2	KTK-R-15A	BUSSMANN	FUSE KTK-R BRANCH CIRCUIT 15 AMP	SEE NOTE 3
1	KTK-R-20A	BUSSMANN	FUSE KTK-R BRANCH CIRCUIT 20 AMP	SEE NOTE 3
2	S501-10-R	EATON	FUSE CERAMIC S501 5x20mm 10 AMP	SEE NOTE 3
2	S501-2-R	EATON	FUSE CERAMIC S501 5x20mm 2 AMP	SEE NOTE 3
2	S501-5-R	EATON	FUSE CERAMIC S501 5x20mm 5 AMP	SEE NOTE 3
2	S501-6.3-R	EATON	FUSE CERAMIC S501 5x20mm 6.3 AMP	SEE NOTE 3

SHOP NOTES:

1. ALL ITEMS ON THIS MODULE (EXCLUDING MANUAL VALVES) INCLUDE THE TAG PREFIX PRD0154.

ENGINEERING NOTES:

1. THIS DRAWING REFERENCES THE ENGINEERING RECORD G02-1-PRDXXXX-E-RIO-01-P01-05 REV 0 FOR ALL TECHNICAL INFORMATION. IN THE EVENT OF DISCREPANCY BETWEEN THIS DRAWING AND THE ENGINEERED DRAWING OF RECORD, THE ENGINEERED DRAWING OF RECORD WILL TAKE PRECEDENCE UNLESS THE CHANGE IS PROPERLY CLOUDED AS A REVISION TO THIS DRAWING AND STAMPED BY A PROFESSIONAL ENGINEER.

NOTES:

1. DISTRIBUTED CHARM IO AND PARTS ON SEPARATED JB
2. SHIP LOOSE
3. SPARE FUSES TO BE PLACED IN UTILITY BOX

REDRAWN FROM SPARTAN
VENDOR DRAWING PACKAGE
DV-54809-ZX-200-00 REV 0

DRAWING NO.	REFERENCE DRAWINGS

STAMPS APPLY TO CURRENT REVISION ONLY

SCOVAN INC.
2026.05.26
ISSUED FOR CONSTRUCTION

REV.	DESCRIPTION	JOB No.	DATE	BY	CHK	PE
1	ISSUED FOR CONSTRUCTION WITH OPTION(S) 3, 9	2026.1111	2026.05.26	AUTO	AUTO	AUTO
0	ISSUED FOR CONSTRUCTION	2026.1111	2026.05.26	AUTO	AUTO	AUTO

PadX PARTNERSHIP
Scovan

PADX
NEXT GENERATION ENERGY SOLUTIONS
PRODUCTION MODULE (MOD-01-P01)
REMOTE I/O PANEL RIO-P01 BILL OF MATERIALS

CLIENT DRAWING NO.	DRAWING NO.	REV.
-	G02-1-PRD0154-E-RIO-01-P01-05	1

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